

Kavindu Munugoda (PhD)

Pharmacognosy | Phytochemistry | Molecular Virology | Natural Product Drug Discovery

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 LinkedIn  Google Scholar  ORCID iD  ResearchGate  GitHub

Summary

Postdoctoral researcher specializing in natural product drug discovery for cancer and viral diseases, with proven expertise bridging traditional ethnobotanical knowledge and modern molecular approaches. Developed WHO-compliant vector control solutions and characterized 127+ bioactive plant species for therapeutic applications.

Research Interests: Natural product-based antiviral drug discovery | HPV therapeutic development | Phytochemical characterization and bioactivity screening | Structure-activity relationship studies | Vector-borne disease control | Ethnopharmacology and traditional medicine validation

Skills

Drug Discovery & Computational Biology

Molecular docking and in silico screening (AutoDock Vina, PyRx, UCSF ChimeraX), protein-ligand interaction analysis, structure-based drug design, structural bioinformatics, genome assembly and annotation

Analytical Chemistry & Natural Products

GC-MS analysis, UV spectroscopy, hydrodistillation, Soxhlet extraction, phytochemical screening and characterization, Micro-Kjeldahl analysis, bioactive compound isolation

Molecular Biology & Protein Biochemistry

Recombinant DNA technology, molecular cloning and plasmid construction, PCR and gel electrophoresis, CTAB genomic DNA extraction, recombinant protein expression (*E. coli*), GST-affinity chromatography, SDS-PAGE, Western blotting

Cell-Based Assays & Virology

Mammalian cell culture across multiple lines (HEK293T, HaCaT, HeLa, CaSki, C33A, SiHa), cytotoxicity profiling and IC₅₀ determination (CCK-8 assay), dual-luciferase reporter systems, cell transfection, viral protein expression and functional characterization

Microbiology & Bioprocessing

Microbial isolation and identification, colony characterization, aseptic technique and subculture, media preparation, fermentation processes, quality control microbiology

Bioinformatics & Data Science

Statistical analysis and modeling (Python, R, GraphPad Prism, IBM SPSS, Minitab), dose-response curve fitting (4PL models), multivariate analysis (PCA, logistic regression, Random Forest), genomic data analysis (NOVOPlasty, MAFFT)

Education

PhD

Pharmacognosy and Phytochemistry

University of Sri Jayewardenepura, Gangodawila, Nugegoda, Sri Lanka

June 2019- July 2023

Dissertation: Production of Mosquito Repellents Using Lesser Known, Under-Utilized Volatile Oil Producing Plant Species

- Identified 127 novel aromatic plant species for mosquito-repellent properties through systematic phytochemical screening and GC-MS analysis
- Developed and field-tested 5 market-ready mosquito repellent formulations following WHO pesticide evaluation schemes and IAEA vector control standards
- Led interdisciplinary research integrating natural product chemistry, ethnobotany, and sustainable biotechnology for rural development applications
- Conducted multi-site community-based field efficacy trials and ethnobotanical knowledge documentation across diverse geographic regions

B.Sc. (Hons) in Applied Sciences | Second Class Upper

University of Sri Jayewardenepura, Gangodawila, Nugegoda, Sri Lanka

January 2015 - December 2018

Major: Plant Biotechnology | Minor: Chemistry and Zoology

Dissertation: Evaluation of Microbial Counts in Apple Flavoured Beer Compared to Base Beer after Exposing to an External Environment

- Identified key beer spoilage organisms (wild yeasts, *Brettanomyces*, *Lactobacillus*, and *Pediococcus* species) through microbiological analysis and standard plating techniques
- Developed evidence-based recommendations for optimized stabilization temperatures in apple-flavored beer production based on microbial susceptibility patterns
- Demonstrated relationship between pH alterations in flavored beer formulations and increased microbial contamination risk

Research Experience

Visiting Scientist | December 2024 – December 2025

International Centre for Genetic Engineering and Biotechnology (ICGEB), China Regional Research Centre, Taizhou City, Jiangsu Province, China

- Conducted *in silico* screening of 51 phytochemicals from 122 Sri Lankan medicinal plants against HPV E1-E2 targets and validated lead compounds through cytotoxicity profiling across mammalian cell lines
- Established dual-luciferase reporter assay system for functional screening of HPV11 E1-E2 replication inhibitors and evaluated 14 combination therapy strategies
- Cloned and expressed HPV11 E1/E2 genes as GST-fusion proteins with optimized IPTG induction protocols and performed affinity purification

Research Assistant | December 2023 - November 2024

Molecular Medicine Unit, Faculty of Medicine, University of Kelaniya, Sri Lanka

- Managed *Aedes* mosquito colony maintenance and rearing protocols supporting vector control and sterile insect technique (SIT) research
- Designed experimental protocols and mentored undergraduate/graduate students in medical entomology research methods
- Contributed to research on dengue vector biology and mosquito-borne disease control strategies

Teaching Experience

Department of Forestry and Environmental Sciences

University of Sri Jayewardenepura, Gangodawila, Nugegoda, Sri Lanka

2019-2023

- Conducted laboratory sessions and lectures in phytochemistry, natural product extraction, and analytical chemistry for undergraduate and graduate students
- Supervised 10+ student research projects on medicinal plants and ethnobotany

Department of Agricultural and Food Technology

University of Vocational Technology, Dehiwala-Mount Lavinia, Sri Lanka

2019-2021

- Delivered lectures on food microbiology, quality control, and fermentation technology
- Mentored students in laboratory techniques and research methodology

Awards & Recognition

SUSRED Award (Support Scheme for Supervision of Research Degrees)

National Science Foundation (NSF), Sri Lanka | 2024

Awarded to honor supervisors whose guidance has produced high-impact doctoral research benefiting the nation.

Best Oral Presentation Award

26th International Forestry and Environment Symposium | January 2022

Session: Environmental Economics and Resource Management

Recognized for outstanding research presentation on "Potential of Extra Income Generation for the Rural Sector by Medicinal Plant Cultivation."

Grants & Fellowships

ICGEB-DIC-MOST International Fellowship Programme (IFP)

ICGEB & Chinese Ministry of Science and Technology | December 2024 – December 2025

Competitive 12-month fellowship for postgraduate research on plant-based antiviral therapeutics against HPV in Prof. Boon Siaw Shi's laboratory, ICGEB China Regional Research Centre.

Patents

Volatile Plant Oil–Based Mosquito Repellent Sticks & Sprays | July 2024

Application No: LK/P/1/23236

National Intellectual Property Office, Sri Lanka

Co-inventors: S. M. C. U. P. Subasinghe; K. D. Munugoda; M. D. Hapugoda

Publications

Published in Peer-Reviewed Journals

- **Munugoda, K. D.**, Thalagala, T.S., Subasinghe, S. M. C. U. P., Hettiarachchi, D. S., Cooray, A. T., & Hapugoda, M. D. (2023). Choice modelling for the commercial cultivation of underutilized aromatic plants for producing mosquito repellents: Targeting rural sector income generation. *Economic Botany* (2023), 1-19. <http://dx.doi.org/10.1007/s12231-023-09573-y>
- Chen, J., Lee, S. Y., **Munugoda, K. D.**, Mohamed, R., Subasinghe, S. M. C. U. P., & Liao, W. (2021). The complete chloroplast genome of Walla Patta, *Gyrinops walla* (Thymelaeaceae), an agarwood-producing tree species from Sri Lanka. *Mitochondrial DNA Part B*, 6(6), 1699-1701. <http://dx.doi.org/10.1080/23802359.2021.1926362>

Manuscripts Under Review

- **Munugoda, K. D.**, Wijesekara, W. L. I., Hettiarachchi, D. S., Rajapaksha, S. M., Subasinghe, S. M. C. U. P., Cooray, A. T., & Jayanika, R. G. I. (2025). Reinventing Artisanal Curry Leaf Paste: Biochemical and Sensory Analysis of Five Sri Lankan Cultivars. *agRxiv*. doi:10.31220/agRxiv.2025.00356 (Preprint)
- **Munugoda, K. D.**, Subasinghe, S. M. C. U. P., Hettiarachchi, D. S., Cooray, A. T., & Hapugoda, M. D. (2024) Refining Nature's Defense: Production and Evaluation of Mosquito Repellents from *Ocimum tenuiflorum* L. (Krishna and Rama Subtypes) and *Ocimum gratissimum* L. Essential Oils doi:10.26434/chemrxiv-2025-5l0v3 (Preprint)

Presentations

Oral Presentations

- Perera M.S.S., Fernando P.J.S., De Silva P.R.S., and **Munugoda K.D.** (2022). Willingness to implement a hydroponics system among Up-country vegetable farmers in Sri Lanka. International Open University Research Sessions (iOURS 2022), Open University, Sri Lanka
- Perera M.S.S., Fernando P.J.S., De Silva P.R.S., and **Munugoda K.D.** (2022). Farmers' Perceptions and Willingness to Adopt Hydroponic Systems: A Study in Welimada, Sri Lanka. International Symposium on Agriculture and Environment (ISAE 2022), Faculty of Agriculture, University of Ruhuna, Sri Lanka
- **Munugoda, K. D.**, Subasinghe, S. M. C. U. P., Hettiarachchi, D. S., Cooray, A. T., & Hapugoda, M. D. (2022). Medicinal aromatic plant cultivation for the bioeconomy: Eliciting willingness from rural communities. Environmental Economics and Resource Management, 26th International Forestry and Environment Symposium 2021, University of Sri Jayewardenepura, Sri Lanka
- **Munugoda, K. D.**, Subasinghe, S. M. C. U. P., Hettiarachchi, D. S., Cooray, A. T., & Hapugoda, M. D. (2020). Potential of extra income generation for the rural Sector by medicinal plant cultivation. Impacts of COVID-19 on Society and Environment, 25th International Forestry and Environment Symposium 2020, University of Sri Jayewardenepura, Sri Lanka

Poster Presentations

- **Munugoda, K. D.**, Buddhinie, P. K. C., & De Silva, C. (2019). Evaluation of microbial counts in apple flavoured beer compared to base beer after exposing to external environment. Uni-In Alliance 2019, University of Sri Jayewardenepura, Sri Lanka

Memberships

Associate Member (A.I.Chem.C.) | Institute of Chemistry Ceylon | Membership No: 3548 Aff

Affiliated to Federation of Asian Chemical Societies (FACS), International Union of Pure and Applied Chemistry (IUPAC), Royal Society of Chemistry, UK and Royal Australian Chemical Institute. <https://ichemc.ac.lk/>

Certifications

PacMOSSI-Aedes & Anopheles Surveillance & Control
James Cook University, Australia
June 2024

Research Methodology and Scientific Writing
Faculty of Graduate Studies, University of Sri Jayewardenepura, Sri Lanka
November 2020

Thermo Fisher Scientific - Mass Spectrometry Job Simulation
Forage
January 2024

Science, Technology and Innovation Policy and Policy Instruments for Sustainable Development Goals for Asia and the Pacific
UN Interagency Task Team on STI for the SDGs (IATT)
November 2024

English

IELTS 7.0



References

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